

A Framework for Finding Statistically Significant Differences

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Abstract

In the questionnaire analysis, finding whether there is a statistically significant difference between two or more groups in a continuous measure is one of the major problems in researches. However, it is difficult for researchers to solve the issue of finding possible statistically significant difference, namely “Statistically Significant Difference Unawareness Issue”. There are two causes to lead to this issue. The first is that the process of finding the statistically significant differences is highly dependent on researchers’ intuition and experience, and the second is that the original questionnaire data may be not good enough to find the statistically significant differences.

In order to solve the Statistically Significant Difference Unawareness Issue, we apply the data warehousing technique and propose a discovery-based framework to help researchers find the possible statistically significant differences. In this framework, three kinds of indicators, Increase, StepDown, and Dice, are designed to explore the data and assist researchers making decisions at different situations.

1 Introduction

Social science research is the use of scientific methods to investigate human behavior and social phenomenon [3][12]. However, the population of human society is generally very huge, and it is impossible for researchers to thoroughly observe the huge population for a behavior or a phenomenon. Hence, they usually use questionnaire survey instead of investigating the whole population.

Questionnaire survey is usually done by selecting some representative samples from population according to the sampling methods. For analyzing the questionnaire data, researchers can use not only descriptive statistics methods to describe the basic distribution of the samples, but

also inferential statistics methods to infer the real human behavior and social phenomenon.

In the questionnaire analysis, finding whether there is a statistically significant difference between two or more groups in a continuous measure is one of the major problems in researches. For example, in a questionnaire survey for elementary school students, “Is there a significant difference between boys and girls about the hours they access the Internet every week?” is an interesting phenomenon that researchers want to know. Barbara Tabachnick [13] categorized the research questions into four types: degree of relationship among variables, significance of group differences, prediction of group membership, and structure, where the significance of group differences is used to find the significant differences. Therefore, finding possible statistically significant differences is a very important research issue.

However, it is difficult for researchers to solve the issue of finding possible statistically significant difference, namely *Statistically Significant Difference Unawareness Issue*, which means that researchers sometimes won’t be aware of existent statistically significant differences in the data. By our observation, there are two main causes to lead to this issue.

The first cause is that the process of finding the statistically significant differences is highly dependent on researchers’ intuition and experience. For example, in a questionnaire survey of elementary school students’ Internet usage behavior, a researcher might consider that there is a significant difference between different genders about the hours they access the Internet every week. She/He could make a hypothesis, “There is a difference between different genders about the hours they access the Internet every week,” and then use inferential statistics method to test this hypothesis. Without making the hypothesis, the statistically significant difference can not be found even if it really exists. However, senior researchers might find it easier because of their rich experiences.

The second cause is that the original

questionnaire data may be not good enough to find the statistically significant differences. For example, in a questionnaire survey of elementary school students' Internet usage behavior, if there is no significant difference between different resident regions about the hours students access the Internet every week, the researcher will consider there is no significant difference between different resident regions. However, each region may contain several counties in geography. Although there is no significant difference between different regions, there might exist between different counties. By combining the secondary data with the original questionnaire data, the student's resident dimension may contain different levels of granularity and she/he could drill down the dimension to find whether there is a significant difference between different counties.

In order to overcome the *Statistically Significant Difference Unawareness Issue*, we first build a data warehouse, which has subject-oriented, integrated, time-variant, and nonvolatile features, to combine the questionnaire data and secondary data, and build frameworks to assist researches finding the possible statistically significant differences from the data warehouse. In this paper, three indicators, *Increase*, *StepDown*, and *Dice*, are designed to help researchers making decisions at different situations. *Increase* shows the degree of statistically significant difference of changing the view by increasing an additional dimension, *StepDown* shows the degree of statistically significant difference of changing the view by stepping down this dimension, and *Dice* shows the degree of statistically significant difference of changing the view by dicing for some values. By using these indicators, users can easily explore the data cube to find possible statistically significant difference.

2 Preliminary

2.1 Quantitative Research

Quantitative research techniques [3][12] are part of primary research and the quantified data can be collected through structured interviews, experiments, or surveys. Quantitative research is all about quantifying relationships between variables. Variables are things like weight, performance, time, and treatment. Researchers measure variables on a sample of subjects, which can be tissues, cells, animals, or humans, and express the relationship between variables using effect statistics, such as correlations, relative frequencies, or differences between means.

"Hypothesis Testing" is the most popular statistics method to analyze the relationship

between variables. And the researchers are most concerned about the differences between means, because it can immediately and effectively indicate the causality of the subject. If some variable like the gender of students versus the hours that students access the internet per week has a difference which is a statistical significance at a given $1-\alpha$ confident level, we said that there is a significant difference between different gender of students.

In the traditional quantitative research, the researchers will firstly propose several hypotheses of the subject according to their experiences, and then determine whether there exists a statistical significance in some hypothesis in a try-and-error manner. The quality of the result using the manner, of course, is in accordance with the quality of the hypotheses made by the researchers.

Furthermore, the quantitative research researches aimed to determine the relationship between one thing and another in a population. Quantitative research designs are either descriptive (subjects usually measured once) or experimental (subjects measured before and after a treatment). An experiment establishes causality. For an accurate estimate of the relationship between variables, a descriptive study usually needs a sample of hundreds or even thousands of subjects; an experiment, especially a crossover, may need only tens of subjects.

2.2 Data Cube

Data warehouse and OLAP [2][4][8] model the data as multidimensional database structure [1][10], and multidimensional database structure views data as data cube [5][6][9]. The data structure of data cube is generally represented in a form of star schema, snowflake schema, or fact constellation schema [7], where the star schema is the most basic one, and the other two can be derived from it. To simplify our discussion, in this paper, a data cube is represented in the form of star schema.

Definition 1. (Data Cube). A data cube is formed by n dimensions, $\langle D^1, D^2, \dots, D^n \rangle$, some measures, and some records. Each dimension D^i contains l_i ordered levels of granularity, called attribute, $\langle A_1^i, A_2^i, \dots, A_{l_i}^i \rangle$. In order to simplify our discussion, assume all the different levels are linear ordered. A_1^i is the most coarse-grained level, and, on the contrary, $A_{l_i}^i$ is the most fine-grained level. The domain of attribute A_j^i is represented as $dom(A_j^i)$, and the data type of each measure is continuous. ■

Example 1. The sample data cube was built based

on a questionnaire survey of elementary school students' Internet usage behavior in Taiwan. This data cube contains 4 dimensions and 1 measure. The detailed schema of this data cube is listed as follows. In the rest of this paper, all the examples are based on this data cube.

- *Dimensions*: $\langle \text{dim_gender}, \text{dim_grade}, \text{dim_resident}, \text{dim_father_edu} \rangle$
 - *dim_gender*: $\langle \text{gender} \rangle$
 - ♦ $\text{dom}(\text{gender}) = \{\text{male}, \text{female}\}$
 - *dim_grade*: $\langle \text{class}, \text{grade} \rangle$
 - ♦ $\text{dom}(\text{class}) = \{\text{lower}, \text{intermediate}, \text{higher}\}$
 - ♦ $\text{dom}(\text{grade}) = \{\text{first}, \text{second}, \dots, \text{sixth}\}$
 - *dim_resident*: $\langle \text{region}, \text{county}, \text{city} \rangle$
 - ♦ $\text{dom}(\text{region}) = \{\text{north}, \text{central}, \text{south}, \text{east}\}$
 - ♦ $\text{dom}(\text{county}) = \{\text{Taipei}, \text{Hsinchu}, \dots, \text{Taitung}\}$
 - ♦ $\text{dom}(\text{city}) = \{\text{Benciao}, \text{Junghe}, \dots, \text{Lanyu}\}$
 - *dim_father_edu*: $\langle \text{father_edu} \rangle$
 - ♦ $\text{dom}(\text{father_edu}) = \{\text{elementary}, \text{junior_high}, \dots, \text{graduate}\}$
- *Measures*: $\{\text{internet_time}\}$

Some notations and operations are used in this paper. To simplify the discussion of notation and increase the readability, we just focus on one specific measurer.

- *Condition*: A 2-tuple of attribute value pair. A condition is used to represent the condition of a dimension. There can be three different types of a condition, including 2 non-universal cases and 1 universal case:
 - (a, v) : The non-universal case that attribute a is with value v , where $v \in \text{dom}(a)$. For example, $(\text{region}, \text{north})$.
 - $(a, *)$: The non-universal case that the attribute a can be any value. For example, $(\text{gender}, *)$.
 - $(*, *)$: The universal case, where any attribute can have any value.
- *Intersection_Condition* ($\wedge$): The

intersection_condition operator is a binary operator that intersects two conditions into a new condition. For example, $(\text{gender}, \text{male}) \wedge (*, *) = (\text{gender}, \text{male})$.

- *Constrain Sequence*: A sequence of n ordered conditions, denoted by $\langle e^1, e^2, \dots, e^n \rangle$, where e^i represents the condition of dimension D^i , a first type of condition, or $e^i = (*, *)$. For example, $\langle (\text{gender}, \text{male}), (\text{grade}, \text{fifth}), (*, *), (*, *) \rangle$ is a constrain sequence.
- *Attribute Sequence*: A sequence of n ordered conditions, denoted by $\langle e^1, e^2, \dots, e^n \rangle$, where e^i represents the condition of dimension D^i , a second type of condition, or $e^i = (*, *)$. For example, $\langle (*, *), (\text{class}, *), (\text{city}, *), (\text{father_edu}, *) \rangle$ is an attribute sequence.
- *All Sequence*: A sequence of the n ordered third type of conditions, i.e., $\langle (*, *), (*, *), \dots, (*, *) \rangle$.
- *Intersection_Sequence* ($\wedge$): The intersection_sequence operator is a binary operator that intersects two sequences to a new sequence.

$$S_1 \wedge S_2 = \langle e_1^1, e_1^2, \dots, e_1^n \rangle \wedge \langle e_2^1, e_2^2, \dots, e_2^n \rangle$$

$$= \langle e_1^1 \wedge e_2^1, e_1^2 \wedge e_2^2, \dots, e_1^n \wedge e_2^n \rangle$$
 , where S_1 and S_2 are sequences. For example, $\langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{north}), (*, *) \rangle \wedge \langle (*, *), (\text{grade}, \text{fifth}), (*, *), (*, *) \rangle = \langle (\text{gender}, \text{male}), (\text{grade}, \text{fifth}), (\text{region}, \text{north}), (*, *) \rangle$.

- *Enumeration* ($\otimes$): The enumeration operator is a binary operator that enumerates a second type of condition with an existent set of constrain sequence into a new set of constrain sequence. Let CSS be a set of constrain sequences. We define $CSS \otimes (a^i, *) = \{CS \wedge \langle (*, *), (*, *), \dots, (a^i, v), \dots, (*, *) \rangle \mid \forall CS \in CSS, \forall v \in \text{dom}(a^i)\}$, where a^i is an attribute of dimension D^i . For example, $\{ \langle (\text{gender}, \text{male}), (*, *), (*, *), (*, *) \rangle, \langle (\text{gender}, \text{female}), (*, *), (*, *), (*, *) \rangle \} \otimes (\text{region}, *) = \{ \langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{north}), (*, *) \rangle, \langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{central}), (*, *) \rangle, \langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{south}), (*, *) \rangle, \langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{east}), (*, *) \rangle, \langle (\text{gender}, \text{female}), (*, *), (\text{region}, \text{north}), (*, *) \rangle, \langle (\text{gender}, \text{female}), (*, *), (\text{region}, \text{central}), (*, *) \rangle, \langle (\text{gender}, \text{female}), (*, *), (\text{region}, \text{south}), (*, *) \rangle, \langle (\text{gender}, \text{female}), (*, *), (\text{region}, \text{east}), (*, *) \rangle \}$.

$\langle (gender, female), (*, *), (region, east), (*, *) \rangle$.

- $CONF = (AS, CS)$: The configuration of current view of the data cube. View will be described in the next section, and the configuration is used to represent the current status of the view. The configuration is a 2-tuple of attribute sequence, AS , and constrain sequence, CS . It means the current view selects all the records satisfying the constrain sequence and observes the attribute sequence. For example, $CONF = (\langle (*, *), (*, *), (region, *), (*, *) \rangle, \langle (gender, male), (*, *), (*, *), (*, *) \rangle)$ means that the current view selects the students who are boys and observes the resident region.
- $\bar{X}$ and S : The mean and standard deviation of all the records.
- $\bar{X}_{cs}$: The mean of the records satisfying CS , where CS is a constrain sequence. For example, $\bar{X}_{\langle (*, *), (*, *), (county, Taipei), (*, *) \rangle}$ is the mean of all the students living in Taipei.

2.3 Indicator

The data warehouse supports the analysis tool OLAP, and users can use some OLAP operations, like roll-up, drill-down, dice, etc., to explore the data cubes. However, the exploring process is not automated, and users still need to explore the data cube by her/his intuition and experience. Sunita Sarawagi [10] proposed a Discovery-driven Exploration of OLAP Data Cubes approach, which provides the following three kinds of precomputed indicators to assist users to explore the data cubes.

- *SelfExp*: This indicates the degree of surprise of the cell value, relative to other cells at the same level of aggregation.
- *InExp*: This indicates the degree of surprise somewhere beneath the cell, if we were to drill down from it.
- *PathExp*: This indicates the degree of surprise for each drill-down path from the cell.

However, these indicators are focused on finding the exceptions in data cubes, not statistically significant difference. In this paper, we use the similar idea of [10] to precompute some indicators to assist users easily finding

statistically significant difference.

3 Idea

In order to overcome the *Statistically Significant Difference Unawareness Issue*, we first build a data warehouse, which has subject-oriented, integrated, time-variant, and nonvolatile features, by combining the questionnaire data and secondary data. Nowadays, researchers generally firstly collect the required data, and then find and learn the appropriate functions to explore and analyze questionnaire by utilizing database, excel, or SPSS software, but the process is very hard. Therefore, we apply data warehousing technology and use OLAP to explore the data online from various views. Although OLAP is easy to explore data, it's not good enough for finding statistically significant differences. Hence, a framework with three kinds of indicator is proposed to assist researchers explore the data cubes in data warehouse and find possible statistically significant differences.

- *Increase*: This shows the degree of statistically significant difference of changing the view by increasing an additional dimension.
- *StepDown*: This shows the degree of statistically significant difference of changing the view by stepping down this dimension.
- *Dice*: This shows the degree of statistically significant difference of changing the view by dicing for some value.

We briefly introduce the idea of this framework and its functionalities (i.e., indicators). When users explore the data cube, they change different views by using the indicators. Each view in the framework shows two different parts as shown in Figure 1: the upper part shows the non-explored dimensions in the data cube, and the lower part shows the detailed information of current view. Each view contains a configuration describing the current status of this view, which is a 2-tuple of attribute sequence and constrain sequence. Figure 1 shows the initial view when exploring the data cube described in Example 1. The upper part shows all the dimensions because they are not explored yet, and the lower part shows the configuration and the mean of the measure *internet_time* of all records.

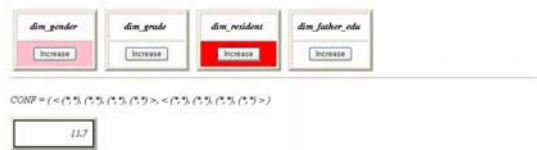


Figure 1. The initial view of exploring the data cube described in Example 1

The different gray-scale color represents the degree of each indicator (In fact, different indicators own different color-scale in the system. However, they all look like gray-scale on the paper.): the darker color represents the higher possibility that there might contain possible statistically significant difference. In Figure 1, the indicator *Increase* of dimension *dim_resident* owns the darkest color, which represents the dimension *dim_resident* has higher possibility existing statistically significant difference. Hence, users can press the *Increase* button of dimension *dim_resident* to explore it, and then the detail of the dimension can be obtained as shown in the lower part of Figure 2.

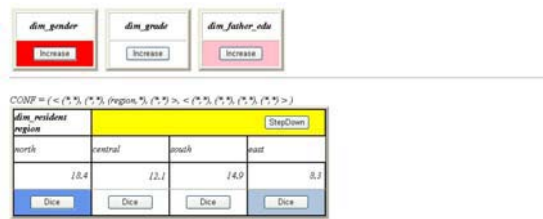


Figure 2. Measure internet_time in different values of attribute region

When users change different views by indicators, the configuration will change correspondingly. After pressing the *Increase* button of dimension *dim_resident* in Figure 1, we could find the configuration also changed from $(\langle (*, *), (*, *), (*, *), (*, *) \rangle, \langle (*, *), (*, *), (*, *) \rangle)$ to $(\langle (*, *), (*, *), (region, *), (*, *) \rangle, \langle (*, *), (*, *), (*, *) \rangle)$ illustrated in Figure 2. It can be easily seen that the higher mean of measure *internet_time* is at north, and the lower mean of it is at east.

At this time, users have three ways to explore the data cube in more detail. First, they can introduce an additional dimension by indicator *Increase*. In Figure 2, the indicator *Increase* of dimension *dim_gender* owns the darkest color among dimensions, and users feel there might contain statistically significant difference. They can introduce the dimension *dim_gender* to the

original view.



Figure 3. Measure internet_time in different values of attributes region and gender

By introducing the dimension *dim_gender* as shown in Figure 3, users can find that the average hours of boys living in north access the Internet is up to 18.9. However, the average hours of girls living in east access the Internet is only 6.9. Hence, there might be statistically significant difference between different resident region and gender about the hours students access the Internet every week. Second, the user can step down the dimension *resident* by indicator *StepDown*. In Figure 2, users can press the *StepDown* button to step down the dimension *resident* from attribute *region* to *county*.



Figure 4. Measure internet_time in different values of attribute county

By stepping down the attribute region to county shown in Figure 4, users can find that the average hours of students living in Taipei access the Internet is up to 20.7. However, the average hours of students living in Taitung access the Internet is only 7.1. Hence, there might be statistically significant difference between different resident counties about the hours students access the Internet every week.

Third, users can dice some value from current view by indicator *Dice*. In Figure 2, users can find the indicator *Dice* of the cell north contains the darkest color. That is, if users select all the students living in north, they have higher possibility to find statistically significant difference. Hence, they can press the *Dice* button

to dice the value *region* = *north*.



Figure 5. Measure internet_time of region = north

In Figure 5, we can find that the configuration changed from $\langle (*, *), (*, *), (region, *) \rangle$, $\langle (*, *), (*, *), (*, *) \rangle$ to $\langle (*, *), (*, *), (*, *) \rangle$, $\langle (*, *), (*, *), (region, north), (*, *) \rangle$. That is, the current view selects all the students living in north of Taiwan and observes whether there is any statistically significant difference on it. There are three different ways to explore the data cube in more detail at each view. By using the three kinds of indicators we design, users can easily find the right path to discover the possible statistically significant difference.

4 Framework

In this section, the mathematical definitions of these indicators are clearly given, and the computational methods of these indicators are also given.

4.1 Score and Range

First, two definitions are given: *Score* and *Range*. *Score* is the z-score in statistics of a group of data, and *Range* is used to evaluate the maximum difference of *Score* between several different groups, that is, the possibility of existing statistically significant difference.

Definition 2. (*Score and Range*). Given a measure, $Score(CS)$ is the z-score in statistic of $\bar{X}_{CS}$, where CS is a constrain sequence. Hence, $Score(CS)$ is defined as:

$$Score(CS) = \frac{\bar{X}_{CS} - \bar{X}}{S}$$

$Range(AS | CS)$ is the maximum difference between all the scores under AS satisfying CS , where AS is an attribute sequence and CS is a constrain sequence. $Range(AS | CS)$ is defined as:

$$Range(AS | CS) = \max\{Score(CS_1 \wedge CS) - Score(CS_2 \wedge CS) \mid CS_1, CS_2 \in \{ \langle (*, *), (*, *), \dots, (*, *) \rangle \otimes (a^1, *) \otimes (a^2, *) \otimes \dots \otimes (a^n, *) \}, (a^i, *) \text{ must be an non-universal element of } AS, i = 1, 2, \dots, n\}$$

$Range(AS | CS)$ can be used to represent the difference between different attribute value combinations in AS under CS . Obviously, the bigger $Range(AS | CS)$ represents the significant difference is more than those of other different attribute values combinations in AS under CS .

Example 2. For the data cube given in Example 1, let $\bar{X} = 13.7$, $S = 4.1$, $\bar{X}_{\langle (*, *), (*, *), (region, north), (*, *) \rangle} = 18.4$, $\bar{X}_{\langle (*, *), (*, *), (region, central), (*, *) \rangle} = 12.1$, $\bar{X}_{\langle (*, *), (*, *), (region, south), (*, *) \rangle} = 14.9$, and $\bar{X}_{\langle (*, *), (*, *), (region, east), (*, *) \rangle} = 8.3$ as shown in Figure 1 and Figure 2. The *Scores* and *Range* can be calculated as:

$$\begin{aligned} & Score(\langle (*, *), (*, *), (region, north), (*, *) \rangle) \\ &= \frac{\bar{X}_{\langle (*, *), (*, *), (region, north), (*, *) \rangle} - \bar{X}}{S} \\ &= \frac{18.4 - 13.7}{4.1} \\ &= 1.146 \end{aligned}$$

$$\begin{aligned} & Score(\langle (*, *), (*, *), (region, central), (*, *) \rangle) \\ &= \frac{\bar{X}_{\langle (*, *), (*, *), (region, central), (*, *) \rangle} - \bar{X}}{S} \\ &= \frac{12.1 - 13.7}{4.1} \\ &= -0.390 \end{aligned}$$

$$\begin{aligned} & Score(\langle (*, *), (*, *), (region, south), (*, *) \rangle) \\ &= \frac{\bar{X}_{\langle (*, *), (*, *), (region, south), (*, *) \rangle} - \bar{X}}{S} \\ &= \frac{14.9 - 13.7}{4.1} \\ &= 0.293 \end{aligned}$$

$$\begin{aligned} & Score(\langle (*, *), (*, *), (region, east), (*, *) \rangle) \\ &= \frac{\bar{X}_{\langle (*, *), (*, *), (region, east), (*, *) \rangle} - \bar{X}}{S} \\ &= \frac{8.3 - 13.7}{4.1} \\ &= -1.317 \end{aligned}$$

$$\begin{aligned}
 & \text{Range}(\langle (*, *), (*, *), (\text{region}, *), (*, *) \rangle, \langle (*, *), (*, *), (*, *), (*, *) \rangle) \\
 &= \max\{\text{Score}(CS_1 \wedge \langle (*, *), (*, *), (*, *), (*, *) \rangle) - \text{Score}(CS_2 \wedge \langle (*, *), (*, *), (*, *), (*, *) \rangle) \\
 & \quad | CS_1, CS_2 \in \{ \langle (*, *), (*, *), (\text{region}, \text{north}), (*, *) \rangle, \langle (*, *), (*, *), (\text{region}, \text{central}), (*, *) \rangle, \\
 & \quad \quad \langle (*, *), (*, *), (\text{region}, \text{south}), (*, *) \rangle, \langle (*, *), (*, *), (\text{region}, \text{east}), (*, *) \rangle \} \\
 &= \text{Score}(\langle (*, *), (*, *), (\text{region}, \text{north}), (*, *) \rangle) - \text{Score}(\langle (*, *), (*, *), (\text{region}, \text{east}), (*, *) \rangle) \\
 &= 1.146 - (-1.317) \\
 &= 2.463
 \end{aligned}$$

4.2 Indicators

Three kinds of indicators are defined in the following:

- *Increase*: This shows the degree of statistically significant difference of changing the view by increasing an additional dimension.

Definition 3. (Increase). Let $CONF = (AS, CS)$ be the configuration of current view. Assume the i -th element of AS is $(*, *)$, i.e., $AS = \langle _, _, \dots, (*, *), \dots, _ \rangle$, which means the i -th dimension isn't explored. The indicator *Increase* is defined as:

$$\text{Increase}_{CONF}(i) = \text{Range}(\langle _, _, \dots, (A_i^i, *), \dots, _ \rangle | CS)$$

After using the indicator *Increase*, the configuration will change from $CONF$ to $CONF'$ via function $f_{\text{Increase}, i}$:

$$CONF' = f_{\text{Increase}, i}(CONF) = (\langle _, _, \dots, (A_i^i, *), \dots, _ \rangle, CS)$$

- *StepDown*: This shows the degree of statistically significant difference of changing the view by stepping down this dimension.

Definition 4. (StepDown). Let $CONF = (AS, CS)$ be the configuration of current view. Assume the i -th element of AS is $(A_j^i, *)$, i.e., $AS = \langle _, _, \dots, (A_j^i, *), \dots, _ \rangle$, which means the i -th dimension is explored to A_j^i . The indicator *StepDown* is defined as:

$$\text{StepDown}_{CONF}(i) = \begin{cases} \text{Range}(\langle _, _, \dots, (A_{j+1}^i, *), \dots, _ \rangle | CS) & \text{if } A_{j+1}^i \text{ exists} \\ 0 & \text{otherwise} \end{cases}$$

After using the indicator *StepDown*, the configuration will change from $CONF$ to $CONF'$ via function $f_{\text{StepDown}, i}$:

$$CONF' = f_{\text{StepDown}, i}(CONF) = (\langle _, _, \dots, (A_{j+1}^i, *), \dots, _ \rangle, CS)$$

- *Dice*: This shows the degree of statistically significant difference of changing the view by dicing for some value.

Definition 5. (Dice). Let $CONF = (AS, CS)$ be the configuration of current view. C is a cell in current view and can be viewed as a constrain sequence. The indicator *Dice* is defined as:

$$\text{Dice}_{CONF}(C) = \max\{\text{Range}(\langle (*, *), (*, *), \dots, (A_i^i, *), \dots, (*, *) \rangle | CS \wedge C) \mid \text{the } i\text{-th element of } AS \text{ is } (*, *)\}$$

After using the indicator *Dice*, the configuration will change from $CONF$ to $CONF'$ via function $f_{\text{Dice}, C}$:

$$CONF' = f_{\text{Dice}, C}(CONF) = (\langle (*, *), (*, *), \dots, (A_i^i, *), \dots, (*, *) \rangle, CS \wedge C)$$

To explain how these indicators are calculated, a real example is given by using the data cube described in Example 1.

Example 3. Figure 6 is a view of the data cube described in Example 1, and the configuration of this view is $CONF = (\langle (*, *), (*, *), (\text{region}, *), (*, *) \rangle, \langle (\text{gender}, \text{male}), (*, *), (*, *), (*, *) \rangle)$.

grade	father_edu
Increase1	Increase2

$$CONF = (\langle (*, *), (*, *), (\text{region}, *), (*, *) \rangle, \langle (\text{gender}, \text{male}), (*, *), (*, *), (*, *) \rangle)$$

resident region				
north	central	south	east	
Mean1	Mean2	Mean3	Mean4	StepDown1
Dice1	Dice2	Dice3	Dice4	

Figure 6. The view in Example 3

The following mathematical formulas are used to explain the calculation formulas of the means and three kinds of indicators by using the definitions described above:

- *StepDown*: This shows the degree of statistically significant difference of changing the view by stepping down this dimension.
- $Mean_1 = \bar{X}_{\langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{north}), (*, *) \rangle}$
- $Mean_2 = \bar{X}_{\langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{central}), (*, *) \rangle}$
- $Mean_3 = \bar{X}_{\langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{south}), (*, *) \rangle}$
- $Mean_4 = \bar{X}_{\langle (\text{gender}, \text{male}), (*, *), (\text{region}, \text{east}), (*, *) \rangle}$

- $Increase_1 =$
 $Range(<(*, *), (grade, *), (region, *), (*, *) > < (gender, male), (*, *), (*, *), (*, *) >)$
 $CONF' = (<(*, *), (grade, *), (region, *), (*, *) >, < (gender, male), (*, *), (*, *), (*, *) >)$
- $Increase_2 =$
 $Range(<(*, *), (region, *), (father_edu, *) > < (gender, male), (*, *), (*, *), (*, *) >)$
 $CONF' = (<(*, *), (*, *), (region, *), (father_edu, *) >, < (gender, male), (*, *), (*, *), (*, *) >)$
- $StepDown_1 =$
 $Range(<(*, *), (*, *), (county, *), (*, *) > < (gender, male), (*, *), (*, *), (*, *) >)$
 $CONF' = (<(*, *), (*, *), (county, *), (*, *) >, < (gender, male), (*, *), (*, *), (*, *) >)$
- $Dice_1 = \max\{$
 $Range(<(*, *), (grade, *), (*, *), (*, *) > < (gender, male), (*, *), (region, north), (*, *) >),$
 $Range(<(*, *), (*, *), (*, *), (father_edu, *) > < (gender, male), (*, *), (region, north), (*, *) >)\}$
 $CONF' = (<(*, *), (*, *), (*, *), (*, *) >, < (gender, male), (*, *), (region, north), (*, *) >)$
- $Dice_2 = \max\{$
 $Range(<(*, *), (grade, *), (*, *), (*, *) > < (gender, male), (*, *), (region, central), (*, *) >),$
 $Range(<(*, *), (*, *), (*, *), (father_edu, *) > < (gender, male), (*, *), (region, central), (*, *) >)\}$
 $CONF' = (<(*, *), (*, *), (*, *), (*, *) >, < (gender, male), (*, *), (region, central), (*, *) >)$
- $Dice_3 = \max\{$
 $Range(<(*, *), (grade, *), (*, *), (*, *) > < (gender, male), (*, *), (region, south), (*, *) >),$
 $Range(<(*, *), (*, *), (*, *), (father_edu, *) > < (gender, male), (*, *), (region, south), (*, *) >)\}$
 $CONF' = (<(*, *), (*, *), (*, *), (*, *) >, < (gender, male), (*, *), (region, south), (*, *) >)$
- $Dice_4 = \max\{$
 $Range(<(*, *), (grade, *), (*, *), (*, *) > < (gender, male), (*, *), (region, east), (*, *) >),$
 $Range(<(*, *), (*, *), (*, *), (father_edu, *) > < (gender, male), (*, *), (region, east), (*, *) >)\}$
 $CONF' = (<(*, *), (*, *), (*, *), (*, *) >, < (gender, male), (*, *), (region, east), (*, *) >)$



The calculation of the indicators contains three steps. The first step is to calculate the $Score(CS)$. Because each the $Score(CS)$ can be calculated by the corresponding aggregation records (each aggregation record is an $\bar{X}_{CS} * N_{CS}$) in the data cube, the $Score(CS)$ is calculated offline when constructing the data cube. The second step is to calculate the $Range(AS | CS)$. Before describing this step, two constants will be defined first. Let n be the size of different combinations of CS , and m be the size of different combinations of AS . Roughly speaking, the size of different combinations of $Range(AS | CS)$ is $n*m$. It will cost a lot of time and space to calculate the total $Range(AS | CS)$. However, since $Range(AS | CS)$ can be easily and fast calculated, the $Range(AS | CS)$ is calculated online. The final step is to calculate the indicators. Because the $Range(AS | CS)$ is calculated when needed, the indicators can be also calculated online.

5 Implementation and Experiment

The prototype of this framework was implemented by using the Analysis Service in SQL Server 2005, which can help us build the data warehouse and provide OLAP tools. Besides, the user interface of this system was written by ASP.NET. When constructing the data cube, an indicator builder module was designed to calculate all the Score values off-line, and save these values into an additional space in data cube. Besides, the Range values and Indicator values were calculated when needed on-line by using the indicator extractor module.

An experiment was designed to evaluate the user satisfaction of this framework by using questionnaire survey. There were 3 parts total 11 questions in this questionnaire: system usage related 5 questions (Q01~Q05), expected functions 4 questions (Q06~Q09), and total evaluation 2 questions (Q10~Q11). Each question was measured by the 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

Two groups of people are involved in the experiment. The first group was composed of senior quantitative researchers, including 2 teachers who taught quantitative research class and 3 senior social science researchers. The second group was composed of 17 students who were studying the quantitative research class. The process of the experiment contains two steps. First, they used the framework by a given data warehouse which will be described later to find all the significant differences in the data warehouse. Second, after using the system, they did the questionnaire.

The given data warehouse was obtained from a questionnaire survey about the current digital divide status of the elementary schools (A project of the Ministry of Education, ROC), and there were total 42,358 samples in this survey. As shown in Figure 7, a digital divide data warehouse was built for this project, and the questionnaire data and some legacy data were integrated into a data cube. Figure 8 shows the star schema of this data cube containing several dimensions with concept hierarchies, like gender, resident, father education, and mother education, and lots of measures, like the “Can you find data on the Internet?”, “Can you collect data with your classmates to finish your homework on the Internet?”, and “How many time do you access the Internet per week?”. The data warehouse was given for these 28 persons, and hoped them find significant differences from this data warehouse. After using the system, a questionnaire survey was taken to measure the satisfying degree of this system.

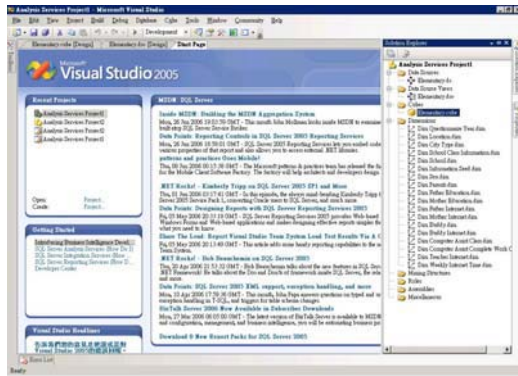


Figure 7. The digital divide data warehouse

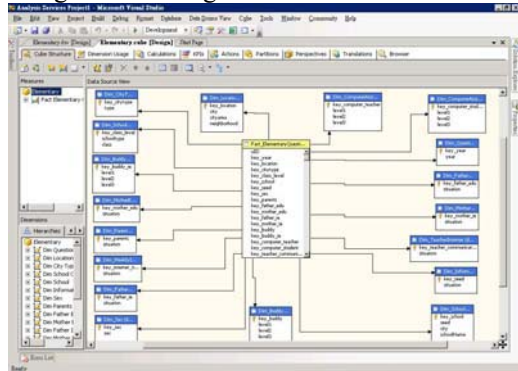


Figure 8. The star schema of the data cube of the digital divide data warehouse

The experiment results are shown in Table 1. The mean points of each question in the first group are lower than the mean points in the second group. The questions with lower mean values (smaller than 3.000) in the first group are Q01 (easy to use), Q02 (friendly user interface), and Q05 (efficiency), and, on the contrary, the questions with higher mean values (greater than 4.000) in the first group are Q03 (speed) and Q07 (hope to automate). On the other hand, the questions with lower mean values (smaller than 3.700) in the second group are Q01 (easy to use) and Q02 (friendly user interface), and the questions with higher mean values (greater than 4.400) in the second group are Q05 (efficiency), Q06 (hope to suggest), and Q07 (hope to automate). Besides, in the comparison of the two groups, there are statistically significant differences in Q02 (friendly user interface), Q04 (correctness), Q05 (efficiency), Q06 (hope to suggest), and Q10 (will use this system again) between two groups.

Table 2. The results of the questionnaire

NO	Question	1st Group		2nd Group		t	Sig.(p)
		Mean	SD	Mean	SD		
Q01	This system is easy to use.	2.800	0.837	3.647	1.115	-1.563	0.134
Q02	The user interface of this system is friendly.	2.400	0.548	3.529	1.281	-2.855*	0.011
Q03	The speed of this system is fast.	4.200	0.447	4.059	0.827	0.362	0.721
Q04	This system can help me find significant difference correctly.	3.000	0.707	4.177	1.185	-2.753*	0.018
Q05	This system can help me find significant difference efficiently.	2.800	0.447	4.412	1.064	-3.257**	0.004
Q06	I hope this system can suggest me the appropriate inferentially statistical methods to test significant difference.	3.400	0.548	4.529	0.943	-2.527*	0.020
Q07	I hope this system can test significant difference automatically.	4.200	0.837	4.647	0.606	-1.334	0.197
Q08	I hope this system can provide the teaching material about the inferentially statistical methods.	3.200	0.837	4.294	1.160	-1.950	0.065
Q09	I hope this system can add different inferentially statistical methods.	3.600	0.548	4.118	0.928	-1.556	0.147
Q10	I will use this system again in the future.	3.000	1.000	4.059	0.966	-2.139*	0.045
Q11	Generally speaking, I am satisfied with this system.	3.400	0.548	4.000	0.791	-1.576	0.131

*p<.05 **p<.01

The mean points of each question in the first group are lower than the mean points in the

second group. That means the acceptability of senior quantitative researchers to the system was lower than the acceptability of junior quantitative

researchers. Since it's a prototype system, Q01 (easy to use) and Q02 (friendly user interface) got the lower points in the two groups: however, this can be improved in the future. There are significant differences in Q04 (correctness), Q05 (efficiency), and Q10 (will use this system again), and the p value of Q05 is smaller than 0.01, especially. That also means the acceptability of senior researchers to the system was lower than the acceptability of junior researchers. Q06 (hope to suggest) got higher points in the second group, and Q07 (hope to automate) got higher points in both groups, which means they are interested in the idea of this system and they hope more useful functions can be added into the system. Generally speaking, they are satisfied with the system.

6 Conclusion

In the questionnaire analysis, finding whether there is a significant difference between two or more groups in one measure is one of the major problems which social science researchers are concerned about. However, finding possible significant differences is difficult for social science researchers. We call it *Significant Difference Unawareness Issue*. In order to overcome the *Significant Difference Unawareness Issue*, in this paper, we firstly build a data warehouse by integrating the questionnaire data and secondary data. Secondly, a framework is proposed to help users find the possible significant differences from the data cube. In this decision support system, three kinds of indicators are designed to assist users.

The prototype of the proposed framework was also implemented, and the experiment about the satisfaction of the system was done. The acceptability of senior quantitative researchers to the system was lower than that of junior quantitative researchers. They were interested in the idea of this system and they hope more useful functions can be added into the system. Generally speaking, they were satisfied with the system.

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